

Python Developer Internship Projects

DecodeLabs Python Developer Training — Batch 2026

Author: Khushal Pa Program: DecodeLabs Python Internship Cohort: Batch 2026 Total Projects: 4 Practical Systems

Project 1 — Interactive To-Do List Application

python todo_list.py

CORE CONCEPTS DEMONSTRATED

Lists append() for loops enumerate() JSON Persistence CRUD Logic

- **Task Creation:** Append new tasks dynamically with default status flags.
- **Status View:** Clean list display with zero-based or 1-based index numbering via `enumerate()`.
- **Task Lifecycle:** Mark individual tasks as completed ("Done") or purge them from the list.
- **File Persistence:** Automatic load and dump synchronization with `tasks.json` on disk.

Project 2 — Personal Expense Tracker

python expense_tracker.py

CORE CONCEPTS DEMONSTRATED

Accumulators while loop try / except ValueError float Type Casting Sentinel Values
Category Breakdown

- **Expense Entry:** Capture numeric amounts, spending categories, and notes.
- **Quick-Add Session:** Continuous continuous-entry mode using loop sentinel values for rapid entry.
- **Accumulator Pattern:** Maintains real-time running balance totals across logging operations.
- **Defensive Inputs:** Guarded against non-numeric entries using robust exception handling.
- **Category Visuals:** Categorized aggregated totals with simple visual ASCII bar breakdown graphs.
- **Persistence:** Preserves transactions in `local_expenses.json`.

Project 3 — Cryptographic Random Password Generator

python
password_generator.py

CORE CONCEPTS DEMONSTRATED

secrets Module string Sets ''.join() Efficiency Entropy Analysis NIST Guidelines
O(N) Complexity

- **Cryptographically Secure:** Powered by the OS entropy source via Python's `secrets` module instead of pseudo-random generators.
- **Guaranteed Diversity:** Enforces strictly balanced inclusion of uppercase, lowercase, numeric digits, and special symbols.
- **NIST Compliance:** Adheres to NIST SP 800-63B standard length limits (8–64 characters).
- **O(N) High Performance:** Assembled via list join methods `''.join()` avoiding costly $O(N^2)$ string concatenation loops.
- **Audit Logging:** Appends generation metadata and logs to `passwords.json`.

Entropy Formula: $E = L \times \log_2(R)$, where L is key length and R is character pool size.

Project 4 — General Knowledge Quiz Game

python quiz_game.py

CORE CONCEPTS DEMONSTRATED

TypedDict Input Validation Modular Functions Set Matching Percentage Scoring
JSON Persistence

- **Interactive CLI UI:** Clean, responsive quiz runner with immediate answer evaluations (✓ / ✗).
- **Type-Safe Structures:** Utilizes strict static typing via Python's `TypedDict` module.
- **Robust Input Validation:** Case-insensitive white-space stripped matching against character sets (`{"A", "B", "C", "D"}`).
- **Grading & Analysis:** Computes percentage performance scores and provides context-aware assessment feedback.
- **Data & State Management:** Reads question banks dynamically from `quiz_data.json` with safe fallback options and replay capability.

Portfolio Project Overview

Project Name	Script File	Data Storage File	Primary Python Focus
1. To-Do List	<code>todo_list.py</code>	<code>tasks.json</code>	Data structures, Iteration, List mutation
2. Expense Tracker	<code>expense_tracker.py</code>	<code>expenses.json</code>	Accumulators, Defensive loops, Error handling
3. Password Generator	<code>password_generator.py</code>	<code>passwords.json</code>	Cryptography, Algorithmic efficiency, Math
4. GK Quiz Game	<code>quiz_game.py</code>	<code>quiz_data.json</code>	TypedDict, Input validation, Grade metrics

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